Lab 8: Templates & Exception Handling

Theory:

- Define templates. Why are templates used?
- What do you mean by exception handling? Why do we use throw, try and catch in exception handling?

Lab:

Question: 1

WAP to show the concept of function template.

```
include<iostream>
using namespace std;
template<class T>
T return_max(T a, T b)
{
T result;
if(a>b)
    result=a;
else
    result= b;
return result;
}
int main( )
{
      cout<<"Between two integer numbers maximum
number="<<return_max(2,3)<<endl;
      cout<<"Between two double numbers maximum
number="<<return_max(5.5,7.5);
      return 0;
}
```

Output:

Question: 2

WAP in C++ to find the sum and average of the elements of the different types of array using function template.

```
#include<iostream>
```

```
using namespace std;
template<class T>
void sum_avg( T a[5])
{
      T sum=0;
      float avg;
      for(int i=0;i<5;i++)
      {
            sum=sum+a[i];
      }
      cout<<"Sum="<<sum;
      avg=float(sum)/5;
      cout<<" Average="<<avg<<endl;
}
int main( )
{
      int a1[5],i;
      double b1[5];
      cout<<"Enter the elements for integer type array:";
      for(i=0;i<5;i++)
      {
            cin>>a1[i];
      }
      cout<<"Sum and Average of integer type array is:"<<endl;
      sum_avg(a1);
      cout<<"Enter the elements for double type array:"<<endl;
      for(i=0;i<5;i++)
      {
            cin>>b1[i];
      }
      cout<<"Sum and Average of double type array is:"<<endl;
      sum_avg(b1);
     return 0;
}
```

Output:

Question: 3

Write a C++ program function template with multiple template parameters.

```
#include<iostream>
using namespace std;
template<class T1,class T2>
void show_data(T1 a,T2 b)
{
      cout<<"("<<a<<" ,"<<b<<")"<<endl;
}
int main( )
{
      int inum=5;
      float fnum=5.5;
      char a[ ]="TEMPLATE";
      show_data(inum, fnum);
      show_data(inum, a);
      show_data(a, fnum);
      return 0;
}
```

Output:

Question: 4

WAP to find the largest element of the array of the template class as well as defining the member function of the template class outside of the class definition.

```
#include<iostream>
using namespace std;
template<class T>
class Array{
      T a[5];
      int i;
      public:
      Array(T a1[5]){
            for(i=0;i<5;i++)
            {
                  a[i]=a1[i];
            }
      }
     void display( )
     {
```

```
      int i;
      for(i=0;i<5;i++)
      {
            cout<<a[i]<<"\t";
      }
T largest( );
};

template<class T>
T Array<T>::largest( )
{
      int i;
      T L=a[0];
      for(i=1;i<5;i++)
      {
      if(a[i]>L)
            L=a[i];
      }
return L;
}
int main( )
{
   int temp1[ ]={1,2,3,4,5};
   double temp2[ ]={1.5,2.5,3.5,4.5,5.5};
   Array<int>obj1(temp1);
   cout<<"The elements of int type array are:"<<endl;
   obj1.display( );
   cout<<endl;
   cout<<"The largest element of the int type array
is:"<<obj1.largest()<<endl;
   Array<double>obj2(temp2);
   cout<<"The elements of double type array are:"<<endl;
   obj2.display( );
   cout<<endl;
```

```
   cout<<"The largest element of the double type array
is:"<<obj2.largest()<<endl;
   return 0;
}
```

Output:

Question: 5

<u>Write a C++ to showing the exception handling.</u>

```
#include<iostream>
using namespace std;
void divide(int x, int y, int z)
{
      if((x-y)!=0)
      {
            int R=z/(x-y);
            cout<<"Result="<<R<<endl;
      }
      else
      {
            throw(x-y);
      }
}
int main( )
{
      int a,b,c;
      cout<<"Enter the three numbers:"<<endl;
      cin>>a>>b>>c;
      try
      {
            divide(a,b,c);
      }
      catch(int)
      {
            cout<<"Exception caught:"<<endl;
```

```
      }
    return 0;
}
```

Output:

Question: 6

WAP in C++ to program that shows handling multiple exceptions.

```
#include<iostream>
using namespace std;
void test( int b)
{
     cout<<"Inside function:"<<endl;
     try
     {
        if(b==0){
          throw b;
          }
         if(b==1)
          {
          throw 1.0;
          }
        if(b==2)
          {
          throw 'R';
          }
     }
     catch(int i)
     {
          cout<<"Integer exception:"<<endl;
     }
     catch(double d)
     {
          cout<<"Double exception:"<<endl;
     }
```

```
    catch(char c)
    {
        cout<<"Character exception:"<<endl;
    }
    cout<<"End of the function:"<<endl;
}
int main( )
{
    test(0);
    test(1);
    test(2);
    test(3);
    cout<<"End of main function:";
    return 0;
}
```

Output:

Conclusion:

- What did you learn from the lab?